Raspberry Pi Laparoscope

Mechatronic Project 478
Proposal

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2026

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[1] Position Statement on Ethical use of Artificial Intelligence in Research and Teaching-Learning Assessment

[2] Draft interim SU guidelines on allowable AI use and academic integrity in assessment

Executive summary

Title of Project
Raspberry Pi Laparoscope
Objectives
The objective of this project is to design, program, build and test a low-cost laparoscope.
What is current practice and what are its limitations?
Current laparoscope use in industry is expensive and not feasible to purchase for training and prototy testing of new laparoscopic medical robots.
What is new in this project?
For this project, instead of specialized lenses and camera modules, off-shelf products will be utilized to develop the product.
If the project is successful, how will it make a difference?
If this project is successful, it can save cost when designing and prototyping new medical robots. It can be utilized for training of new surgeons trainees.
What are the risks to the project being a success? Why is it expected to be successful?
The main risk of this project is delayed due to unavailability of hardware components. Therefore, the ordering of products is the main priority during early stages of the project.
What contributions have/will other students made/make?
Another student is performing the same project is me, thus notes and methods will be compared.
Which aspects of the project will carry on after completion and why?
The developed laparoscope will be utilized in future projects, where medical robots will be devoloped. The test station will also be utilized in testing future laparoscope designs.
What arrangements have been/will be made to expedite continuation?
To prevent delays, safety measures have been taken to prioritise the ordering of important hardware components.

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Chapter 1

Introduction

1.1 Background

The Hippocratic Oath is a fundamental principle that guides decisions and advancements in the medical profession. It motivates doctors and physicians to make ethical decisions to minimize patient harm. It also drives physicians to continuously improve practices to reduce surgical risks and complications. This principle has driven the discovery of minimally invasive surgery (MIS). (Ibrahim *et al.*, 2021)

Minimally invasive surgery has become the standard practice across surgical fields and is a rapidly growing division in the medical industry. One branch of this field is laparoscopy, which is used to perform advanced and complex surgical procedures. (Ibrahim *et al.*, 2021)

Laparoscopy is surgical approach where small incisions are made to allow insertion of specialized equipment into the patient's body. The primary instrument is the laparoscope, which is typically paired with other thin rod equipment based on the operations demands. Compared to traditional open surgery, laparoscopy offers advantages including faster recovery rates and reduced morbidity and patient discomfort, though it requires a steeper learning curve. These benefits translate to cost savings through shorter hospital stays and reduced analgesic requirements. (Subido *et al.*, 2018)

The laparoscope is a fundamental instrument to laparoscopic surgery. It consists of a thin rod with an attached camera that is inserted into the patient's body. The captured footage is displayed in real-time to the surgical staff on a monitor. Based on the visible footage, the surgical staff make diagnosis, perform procedures, and coordinate instrument positioning based on the visual feedback. (Ibrahim *et al.*, 2021)

Recent advancements in laparoscopy focus on improving the laparoscope's vi-

sual capabilities through several approaches. These approaches include increasing the camera's field of view, implementing three-dimensional (3D) visualization, and incorporating artificial intelligence (AI) or machine vision technologies to captured footage. Additional advancement made to modern laparoscopes is the utilization of robotic manipulators, with 3 or more degrees of freedom (DOF) to assist the doctor during operations. (Subido *et al.*, 2018)

One of the main challenges of laparoscopic surgery is that it is a technically complex surgery to perform. Thus, more training is required by inexperienced surgeons. However, laparoscopes are expensive equipment which makes them inaccessible to individual practitioners or small hospitals to purchase. (Xiao *et al.*, 2014)

Therefore, there is a need of a low-cost laparoscope.

1.2 Objectives

The aim of this project is to design, build and test a low-cost laparoscope.

This project's main objectives are to:

- Design, build and program a functional laparoscope by incorporating various Raspberry Pi components.
- Design a performance test to assess the laparoscope functionality and image quality abilities.
- Evaluate the developed laparoscope performance after performing the test.

1.3 Scope

To develop the laparoscope, there is a various components that needs to be considered. The main 3 components are that of design, hardware and software. And each has component has their own set of scopes that need to be considered. These scopes are subject to change based on the timeline of the project and change of desired objectives of the project.

1.3.1 Design Component

For the design component of the project, the only thing that is required is to design a functional laparoscope. The end product should look and function like a medical-grade laparoscope, however, it does not require to be built out of

medical-grade material, since it would not be used on actual patients. All hardware components should be secure and covered within the design to prevent any damage.

The viewing angle, the angle between the optical axis of the camera and the laparoscope axial axis, should aimed to be 30 degrees, as per standard laparoscopes. The viewing angle does not require to be changeable during or between operations. Consequently, no degrees of freedom are required for the final design of the product.

To determine the developed product capabilities, a test is required. A physical box trainer is an ideal product to determine the laparoscope's capability. The mechanical and mechatronic department in Stellenbosch University, however, does not have this product available. Therefore, a physical box trainer or similar product has to be design and build to determine and demonstrate the developed laparoscope's functionality. This design and manufacturing of the testing station is within the project's scope.

1.3.2 Hardware Component

The hardware component is the critical for this project's success. Therefore, it is important to determine what is within the project's scope for this section. For acquiring the various hardware modules, it is not required to design the physical components, themselves. Rather, available products will be purchased to fulfil any hardware required for the project. Hence, proper analysis of available products is required to fulfil the main objectives and satisfy the desired outcomes of the projects. However, if the need comes forth, certain hardware modules could require modification, based on the products that is purchased. These modifications does fall into the project's scope.

It is also within the projects scopes to determine how all of the various different modules are interconnected. These connection can vary from physical to wireless connectivity, but proper consideration should be given when designing the system. Power supply, illumination and circuit design is within the project's scope and should be given proper consideration.

1.3.3 Software Component

The software component of this project is a subject to change, based on the outcomes of the visual capabilities of the laparoscope during testing phases. Currently, there is 3 main categories of the software component: Visual, connectivity and miscellaneous.

The visual category refers to program required to operate the camera module

of the laparoscope. This includes signal processing and real-time digital image enhancement. The real-time digital image enhancement, however, has it own scope. It is only required to enhance the image to determine the laparoscopes visual capabilities. Therefore, it is not required to extract information from the real-time images or incorporate Artificial Intelligence or 3D visualization to the program.

To determine the laparoscopes real-time visual capabilities, it is required to display the real-time images on a monitor . As a result, the laparoscope's program should communicate with the monitor either through a cable or wireless connection, based on the hardware decision. Henceforth, the program should incorporate the communication between the 2 or more devices.

The miscellaneous section of the program is small features that adds value to the developed laparoscope. These small features includes function that stores the captured footage on a external storage or utilizes the pins of Raspberry Pi module to indicate status signals of the laparoscope. These features are not critical to the project, but adds value to the end product.

1.4 Motivation

As mentioned in the *Background*, the main challenge with laparoscopic surgery is that it is a technically more complex procedure to perform, compared to open surgery. Thus, inexperienced surgeons require more training with the laparoscope and its additional equipment to achieve proficiency. However, laparoscopes are expensive surgical equipment, that isn't affordable investment for individuals or small medical facilities.

This project could also add value to the development of new medical robots. Most preliminary designs of medical robots require laparoscopes that can test the robot's capabilities and flaws. However, they do not require expensive medical-grade laparoscopes, rather, low-cost alternatives are preferable, as they minimize financial loss if damaged during testing or development.

This project can also prove if low-cost medical-grade laparoscopes with inexpensive hardware components are achievable for surgical operations. For resource limited settings, this can be extremely valuable due to it improving the healthcare of these settings. It would also enhance the trust of the community in its local medical services.

By developing a low-cost, functional laparoscope, this projects contributes to improved surgical training accessibility, supports the development of medical robots, and demonstrates the feasibility of affordable medical technology for low-resource medical facilities.

1.5 Use of AI Tools

Anthropic is the main Artificial Intelligence (AI) company that will be used for this project, specifically its Claude Sonnet 4.5 model. This AI model assisted with research papers identification, sentence construction, grammar checks and code debugging. All AI-generated material will be reviewed and verified by the student to ensure the integrity and originality of the provided work.

1.6 Proposal Overview

This proposal aims to enlighten the reader of the various sections of work required to perform this project. The *Literature Review* chapter informs of the history of laparoscopy, afterwards it explains various segments of the work required to understand to complete the project. The *Content Chapter* aims to provide an overview of the entirety of the project work procedures. It describes the research methodology utilized; the workplan; the resources required; the expected results; the schedule, including a gantt chart; and the risks associated with this project. The last chapter of this proposal, *Conclusion*, provides a final motivation why this project should be performed.

Chapter 2

Literature Review

A literature review has been conducted for this project. This section of the report documents the relevant research performed and divides into to the following sections:

- **Laparoscopy Overview:** An overview of what laparoscopy is and its history.
- **Current Technological Advancements:** The current directions that modern laparoscopy is improving on.
- **Current Available Technology:** A summary of current Laparoscopes that is available in South Africa.
- **Available Hardware Components:** An overview of different hardware modules available for this project.
- **Software Considerations:** An brief overview of the available programs that can be used for this project.
- **Visual Inspection Considerations:** An overview of visual inspection and aspects of camera that need to be mastered.

2.1 Laparoscopy Overview

Laparoscopy is a branch of the endoscopic medical field. It has become an integral part of surgical procedures in all field in the recent decades. Inspired by the minimally invasive surgery (MIS) approach, laparoscopic surgery consist of making 1 or more small incisions on the abdomen of the patient. Surgeons use these incision to insert thin rod medical equipment into the patient to perform the surgery. (Ibrahim *et al.*, 2021)

An essential tool of this surgical procedure is the laparoscope. Due to small size of the incisions, the surgeons' view is limited. The laparoscope is the solution to this problem. It allows the surgeons to view what is occurring within

the patient while the operation is being performed. Therefore, they can clearly manipulate their equipment from outside the patient to perform complex procedures inside the patient. (Subido *et al.*, 2018)

Modern laparoscopes are equipped with small cameras and light sources at the end of the tool. This camera is inserted into the patient and the captured footage is displayed in real-time to the surgical staff with the means of a monitor. With the visible footage, the surgical staff can make diagnosis and perform operations. (Subido *et al.*, 2018)

However, the first laparoscope has existed for almost the last century. They didn't consist of cameras, rather, out of complex lens assemblies at the end of the laparoscope and cold light sources. They were used in a similar method as a microscope. However, this severely limited the surgeon's ability to operate as well as limited the surgical staff ability to assist the head surgeon.(Ibrahim *et al.*, 2021) *Figure 1* illustrates how the original laparoscope was designed.

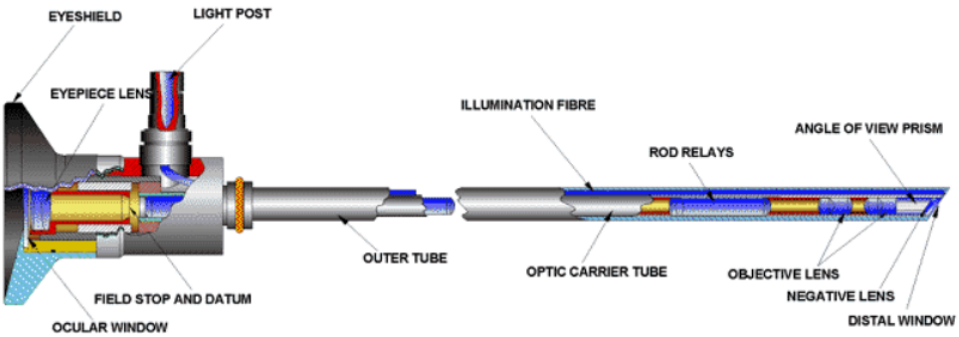

Figure 2.1: Original laparoscope design. (ebme.co.uk, 2026)

Laparoscopic surgery has been one of the greatest advancements in medical industry in recent years. *Buia* et al. *(2015)* shows that laparoscopic surgery has quicker recovery time and less pain after surgery compared to traditional open surgery. However, it shows that certain procedures are more time-consuming and more complex to perform.

2.2 Current Technological Advancements

Since the addition of cameras to laparoscopes, there has been a large addition of advancement available. These advancement range from hardware improvements to additional software programming abilities to the existing design.

The main hardware advancement made to the camera module is increasing the field of view (FoV). However, there are additional advancements made in hardware for laparoscopes. The addition of robotics to the laparoscopes has greatly

assisted the laparoscopic surgical industry. Robotic manipulators are utilized to assist surgeons by maintaining stable control of the laparoscope, therefore, ensuring consistent and clear visualization during the procedure.(Subido *et al.*, 2018)

There are multiple additions currently being made to the software of laparoscopes. *Subido* et al. *(2018)* list the following additions:

- Digital imaging enhancement techniques.
- AI incorporation.
- 3D visualization.
- Object tracking.

2.3 Current Available Technology

There is a range of different laparoscope or endoscope suppliers. Each of these suppliers produce more than 1 product. Therefore, only suppliers for South African hospitals will be reviewed and Intuitive Surgical that produces the *Da Vinci* system, that has been dominating the surgical robot market. (Subido *et al.*, 2018)

One of the main suppliers of laparoscopic equipment in South Africa is Karl Storz Endoskope, a German company, with a sales office in Cape Town. They produce various products in the medical industry. (karlstorz.com, 2026*a*) One of their products is the *TIPCAM1 RUBINA*.

TIPCAM1 RUBINA is a rigid endoscope with the ability of 0° and 30° viewing angle. It also comes with automatic horizon control and the ability to easily swap between 2D and 3D imaging. Its has an outer diameter of 10.3 mm and a working length of 320 mm. It is has 4K image resolution and has a mass of 420 g. (karlstorz.com, 2026*b*)

Figure 2.2: Figure of the TIPCAM1 RUBINA. (karlstorz.com, 2026*b*)

Olympus Corporation is another company that supplies medical equipment globally. They are a Japanese manufacturing company that produces the *ENDOEYE 3D 10 mm*. The *ENDOEYE 3D 10 mm* is a video telescope, with the ability of 3D imaging at a viewing angle of 0° and 30°. It is able to perform image rotation at 30° viewing angle, while maintaining the horizon. (olympus.com.au, 2026)

Figure 2.3: Figure of the ENDOEYE 3D 10 mm. (olympus.com.au, 2026)

Certain international companies uses local distributors to sell their products. Intuitive Surgery, an American company, uses this method. They designed and manufacture the *Da Vinci* medical robot. This system has various variants for different surgical purposes. Currently, there is *Da Vinci Xi* and *Da Vinci X* for multi port surgery. However, the *Da Vinci SP* is designed for single port surgery. These robot systems are are equipped with 7 DOF movement and 3D visualization. Their only drawback is there large size and enormous financial procurement cost. (intuitive.com, 2026)Below is a figure of the *Da Vinci SP*.

Figure 2.4: Figure of the Da Vinci SP. (intuitive.com, 2026)

2.4 Available Hardware Components

There multiple hardware components required to build a laparoscope. Therefore, this section will be divided into multiple subsection where different products will be reviewed for each component. These subsections include:

- Single board computer module.
- Camera module.

2.4.1 Single Board Computer Module

Single board computers (SBC) is a credit-card-size computer, able to perform complex computations while operating various systems. For the laparoscope, it will be the main "brain" of the system. Therefore, it is critical that it can perform all necessary tasks it requires. There is 2 main suppliers of SBC:

- Raspberry Pi Ltd.
- Arduino AG.

Raspberry Pi has a large range of products that is available for the project. However, Raspberry Pi's SBCs has an important feature installed into most of SBC, that would be beneficial to the project. This feature is the designated camera serial interface (CSI) or the MIPI camera display transceivers, according to Raspberry Pi. This feature allows for high-quality videos at high speeds. (raspberrypi.com, 2026*a*) The *table* below show the available options of all the Raspberry Pi SBC:

Table 2.1: Raspberry Pi SBCs.(raspberrypi.com, 2026*b*)

Raspberry Pi SBC	RAM	CPU Speed	MIPI Ports	Cost
Pi 5	1-16 GB	2.4 GHz	2	R 3 266.00
Pi 4 Model B	1-8 GB	1.5 GHz	1	R 1 360.00
Pi 3 Model B+	1 GB	1.4 GHz	1	R 640.43
Pi Zero 2 W	512 MB	1 GHz	2	R 240.16
Pi Zero 2	512 MB	1 GHz	2	R 309.90

All of the SBC in *table* above have these features:

- 40 GPIO Pins.
- Wi-Fi Capabilities.
- Bluetooth Capabilities.

Arduino, similar to Raspberry Pi, have a wide range of products available for this project. However, they have one absent feature, namely the lack of CSI feature. Therefore, they are incompatible with the Raspberry Pi Camera Modules. However, they can use other camera modules, with the use of SPI and I2C GPIO pins. Although this options isn't favourable, due to trade-off in quality during longer videos, it is still a feasible option that can be explored. (Public, 2022) The *table* below shows the available products from Arduino:

Table 2.2: Arduino SBCs. (store.arduino.cc, 2026)

Arduino SBC	RAM	CPU Speed	SPI Ports	Cost
UNO R4 WiFi	32 kB	48 MHz	1	R 488.46
UNO R4 Minima	32 kB	48 MHz	1	R 352.33
UNO R3	2 kB	16 MHz	1	R 469.24
GIGA R1 WiFi	1 MB	480 MHz	2	R 1 247.58
Nano ESP32	512 kB	240 MHz	1	R 358.74
Nano RP2040 Connect	264 kB	133 MHz	4	R 389.17
UNO Q	2 GB	2 GHz	1	R 1 035.54
Nano 33 BLE	256 kB	64 MHz	4	R 456.43
Nano 33 IoT	520 kB	240 MHz	6	R 427.6
Nano Every	6 kB	20 MHz	1	R 232.22

2.4.2 Camera Module

The camera module is another important component of this project. It is essential to capture the real-time images inside the patient. Therefore, this component should be selected carefully. A common occurrence with the camera modules is that the size of the camera sensor and assembly can fit in a Ø10 mm hole. However, the circuit board that it is attached to it does not fit inside this hole. Therefore, options with removable camera assemblies that can be attached to the circuit board with a wire connection were reviewed. Consequently, camera modules that already fit inside with the hole, that comes with a wire extension is reviewed as well.

Raspberry Pi camera modules all have all circuit board connection, that can't separate from the camera sensor, as can be seen in the *figure* below. However, it is optional to purchase the stand-alone sensor assembly for the Raspberry Pi camera module V3. Although, this camera sensors it powerful, with IMX708 sensor capable of 4608 x 2592 pixels images, it physical size is 10.8 mm x 10.8 mm. Therefore, it is incapable to fit inside the Ø10 mm hole. (arducam.com, 2020)

As mention in the previous section, *Single board computer module*, the Rasp-

Figure 2.5: Figure of Raspberry Pi Camera Module 3

berry Pi SBC modules have CSI ports. Even though Raspberry Pi camera modules can't be considered for this project, other suppliers make camera modules that can interface with these CSI ports. Below is a *table* of the supplier, Leopard Imaging, from their hummingbird mobile camera modules series:

Table 2.3: Leopard Imaging camera modules.(leopardimaging.com, 2026)

Product code	Resolution	Max fps	FoV (°)	Focus depth
LI-OV2685-MIPI-FF	2 MP	120	60	30 cm - inf
LI-OV8865-MIPI-AF	8 MP	60	70	10 cm - inf
LI-OV13850-MIPI-AF	13 MP	30	70	10 cm - inf
LI-OV5640-MIPI-AF	5 MP	120	65	10 cm - inf
LI-AR1335C-MIPI-061H-AF	13 MP	30	61	10 cm - inf
LI-IMX258-MIPI-AF	13 MP	60	79.8	10 cm - inf
LI-IMX586-MIPI-AF-79D	48 MP	240	79	N/A
LI-IMX681-MIPI-AF-96D	12 MP	30	96	N/A
LI-AR0246-FLEX-80H	2 MP	60	80	35 cm - inf
LI-OV16E10-AF-D78	16 MP	120	78.4	7 cm - inf
LI-AR0544-MIPI-150H	5 MP	60	150	15 cm - inf
LI-OS04C10-MIPI-107H	4 MP	240	107	70 cm - inf
LI-OG12B10-MIPI-AF-95D	12 MP	240	75	10 cm - inf
LI-OV13B10-MIPI-AF-75D	13 MP	60	75	10 cm - inf

Another camera module manufacturer is ArduCam Technology Co., Ltd. They specialize in embedded camera modules and is a leading company in the designing of camera modules. They also produce several camera modules, however,

their most relevant product to this project is their *OV5647 Spy Camera module*. Here is a list of the products specifications:

- Resolution: 5 MP.
- Max fps: 90.
- FoV: 120°.
- Cost: R 316.89

This product however does have one flaw. Its focus depth is 1 m to inf.(store.arduino.cc, 2026) Therefore, this product cannot be used for this project.

Figure 2.6: Figure of Arducam spy camera module.(store.arduino.cc, 2026)

For this project, there is an option of purchasing endoscope camera modules. These camera modules are designed to be used in a endoscopic application. There are various designers and manufacturers and each of these have their own series of products.

IADIY Technology is a Chinese company, that creates a line of products that can communicate with USB cable attached to the camera module. Below is a *figure* of a IADIY camera module. These camera modules are powerful, with their compact size, wide Field-of-View angles and built in light sources. Here is a *table* that list all the available products from IADIY technology

Figure 2.7: Figure of IADIY camera module.(iadiy.com, 2026)

Table 2.4: IADIY endoscopic camera modules. (iadiy.com, 2026)

Product Code	Resolution	Ø of Module (mm)	FoV (°)	Max fps
SCM03M30D5.5U	0.3 MP	5.5	60	30
SCM1M30D5.5U	1 MP	5.5	88	30
SCM1M30D3.9U	1 MP	3.9	140	30
SCM2M30D8U	2 MP	8	80	30

Guangzhou Sincere Information Technology Ltd. (SincereFirst is their trade name) is another Chinese company. They specialize in various camera products that can be used through USB connection. (cameramodule.com, 2026) Below is a *table* of all the available products from SincereFirst.

Table 2.5: SincereFirst endoscopic camera modules(sincerefirst.com, 2026)

Image sensor	Resolution	Ø of Module (mm)	FoV (°)	Max fps
OmniVision OCHFA20	720 P	1.05	120	30
OmniVision OCHFA10	0.5 MP	1.5	105	30
N/A	0.5 MP	2	120	30
Brigates ES101	1 MP	3.1	140	75

Motoshot Technology Co., Ltd. (Camemake is their trading name) is a Chinese company. Since their establishment in 2016, they have been designing and manufacturing camera modules that can interface with USB, DVP/SPI and CSI. They have 3 endoscopic products that can be considered for this project, as well as a range of other camera module.(camemake.eu, 2026*b*) Below is a *table* that shows some of the products specifications:

Table 2.6: Camemake endoscopic camera modules. (camemake.eu, 2026*a*)

Sensor	Resolution	Ø of Module (mm)	FoV (°)	Max fps	Focus Depth
N/A	1080 P	7	60	30	30 - 50 mm
OV2640	2 MP	1.6	120	30	5 - 50 mm
OV9734	720 P	4.2	100	30	5 -100 mm

All of these products mentioned above in endoscopic camera module tables, all have built in illumination modules. Thus, minimizing the amount of work required to design and manufacture the developed laparoscope.

2.5 Software Considerations

This project is mainly a design problem that has to be addressed. There are however software application needed for this project. The most important ap-

plication is that of the camera module interface.

The software should enable the SBC to connect and operate the camera module. raspberrypi.com (2026*a*) properly documents how to use a camera module that interfaces with an CSI connection. If a SPI/I2C connection is required to interface with the camera module, Public (2022) is library that describes how it can be performed. If a USB connection is used as an interface between the SBC and camera module, then minimal action is required to interface the 2 modules. (mechatronicslab.net, 2026)

There are however other applications required for this project. As mentioned in the *Software Component* of the *Scope*, there are 2 other section of code that need to be considered for this project. The ability display the captured footage to a real-time monitor, depends on the monitor that will be used for this project. If a personal computer is used, the captured footage can be transferred with the means of WiFi, Bluetooth, USB connection, or even a Mathworks ThingSpeak channel.(uk.mathworks.com, 2021)

Another important aspect to considered is coding language used for this project. However, this is dependent of the SBC used. If a Raspberry Pi is used, then Python is recommended language to use. (raspberrypi.com, 2026*a*) However, if Arduino is used then the ideal language to use is either C or C++. (docs.arduino.cc, 2026) Visual Studio Code (VS Code) can be used to perform all programming for this project.

Documentation of this project will be performed using LaTeX. VS Code will be used to run this application, through the LaTeX Workshop extension.(Yu, 2026)

2.6 Visual Inspection Considerations

This project requires a visual camera to be a functional laparoscope. However, there is various factors to be considered to take high-quality images. This section of the literature reviews aims to explain each of these factors. Consequently, how they can impact this project.

Important concept to understand, is how a digital image is captured. A camera sensors consists of a matrix of millions of light cavities. When a picture is being taken, these cavities, called ”photosites”, are exposed to the photons of the object or scene. The photons are then collected by the photosites and stored as an electrical charge. Consequently, when the image is finished being taken, these charges are quantified and then processed to make the image. (cambridgeincolour.com, 2020*b*)

One aspect that needs to be mastered to captured quality images is the exposure

triangle. It is 3 pillars that dictates the quality of an image. Aperture determines the size of the area opened in the lens which light can filter through to sensor. The larger the aperture value, measured in f-stop values, the smaller the hole size where light filters through the lens. This pillars relates to the depth of field of the images. The smaller aperture correlates to shallower depth of field images. (cambridgeincolour.com, 2020*a*)

Shutter speed is the second pillar to the exposure triangle. When a digital images is being taken the lens opens up to expose the photosites to the light. Consequently, shutter speed refers to total duration of the lens being open, thus causing the sensor to be exposed to image. If the shutter speed increases, then the sensors is exposed to reduced light from the images. Therefore, as the shutter speed increases, the images will gain more quality, with more definitive objects that is more "frozen" in the image. (cambridgeincolour.com, 2020*a*)

The third pillar is ISO. This determines how sensitive the sensor is to the exposed light. As the ISO value increases, the higher the sensitivity value will become. Therefore, the image will have increased levels of noise captured if the ISO value is increased. (cambridgeincolour.com, 2020*a*)

Focal length is a critical feature that has to be considered for this project. Focal length, is the distance between the camera sensor and the camera lens. As this value increases, then the field of view value will decrease. Field of view, measured in degrees, is essentially how much of an images can be seen. If this value increases, then a wider image can be taken. (Morrow, 2026)

Chapter 3

Content Chapter

3.1 Research Methodology

Research methodology refers to the structured approach a researcher utilizes to explain and solve a research problem. For this project, a design research methodology will be utilized to solve the problem.

The problem for this skripsie, is to create a low-budget laparoscope. This laparoscope will be used in tests, for medical robots, and not on alive patients, rather, it will be used on phantoms. Therefore, the laparoscope should not be made out of medical grade equipment, but should functional similar to a medical laparoscope.

Since this project requires a design research methodology to be completed, there is a structure that can be followed. The initial phase that needs to be completed is a design requirement phase. In this section it required to determine:

- Stakeholders;
- Design independent parameters;
- Boundary between wider system of interest and narrower system of interest;
- Functional Requirement;
- Non-functional Requirements;
- Technical performance measures;

of the system. It is essential to remain neutral in this section.

The second phase that needs to be completed is the concept development phase. In this section, concepts are developed and described to meet all the requirement in the design requirement phase. These concepts are evaluated before the

best concept or combination of concepts is selected. The selected concept(s) is refined further, before a final evaluation is performed.

The last 2 sections of this methodology is the detail design and analysis phase and the evaluation phase. The detail design and analysis phase requires the researcher to utilize the refined concept from the preceding section and create a detail design of the concept. This process does require an iterative process to create the final design. A drawing pack is required at the end of this phase that will be evaluated. The evaluation phase consist of evaluating the developed product. This includes building the product, before it can be tested. Based on the test results of the product, an evaluation can be executed.

3.2 Workplan

This section of the proposal covers the planned workload that will be conducted for this project. As established in the prior section, *Research Methodology*, this project will follow a design research methodology. Therefore, it can be assumed that the project will require these following activities:

3.2.1 Literature Review

Study the literature of laparoscopic surgery to gain an understanding of the conditions that a laparoscope will experience. Consequently, there will be required to study literature that will inform what attributes the developed laparoscope should satisfy, to be used for testing purposes on future projects.

3.2.2 Component Review

Study and identify the components that can be utilized for this project. The developed laparoscope is dependent on the specifications of the acquired hardware. Therefore, a review is necessary to acquire the highest-performing components at reasonable cost.

3.2.3 Compile Design Requirements for Laparoscope

Determine the requirements for the developed laparoscope. Specify a set of evaluation criteria to determine the best concept during the concept generation phase.

3.2.4 Generate Concept Design for Laparoscope

Describe, formulate and design multiple concepts to meet the design requirements. The derived concepts is to be compared and evaluated. Afterwards the

highest performing concept, or combination of concept, will be selected for a refinement and final evaluation.

3.2.5 Finalise Detail Design for Laparoscope

Design and finalise the selected concept design of the laparoscope. This includes producing relevant calculations and analysis of the developed laparoscope and the full drawing pack of the laparoscope.

3.2.6 Compile Design Requirements for Test Station

Determine the requirements for test station of the laparoscope. These requirements include a list of functionality test that should be performable on the test station.

3.2.7 Generate Concept Design for Test Station

Describe, formulate, articulate and design the various tests required for the test station. These tests are to be evaluated and selected based on threshold value they should meet. All the selected tests will then be implemented in a final test station concept, which upon completion would be evaluated.

3.2.8 Finalise Detail Design for Test Station

Articulate and design the final test station for the laparoscope. This includes producing a safety report of the test station, instruction documents of how the various test should be performed and a full drawing pack of the test station.

3.2.9 Construct and Implement Laparoscope

Construction of the developed laparoscope should be conducted. This includes all software implementation and building of the developed product.

3.2.10 Construct Test Station

Construct the test station for the laparoscope. This entails all building requirements of the test station.

3.2.11 Perform Functionality Test

Perform the functionality tests of the built laparoscope. This includes following all test instructions and post processing of the results.

3.2.12 Evaluate Laparoscope

Evaluate the performance of the developed laparoscope.

3.2.13 Documentation

Document the entirety of the project. All relevant literature review, design specifications, concept designs, test procedures, results and evaluation should be thoroughly documented. At predetermined dates, it is required to produced relevant documents, including the progress report, preliminary report and final report.

3.3 Resource Requirements

There are various equipment, components and lab access required to complete this project. As mentioned in the literature review's section, *Available Hardware Components*, there are 2 main hardware components required for this project. However, there are various hardware components required for this project:

- Single board computers.
- Camera Module.
- Display screen or monitor.
- Power supply.
- Electrical components.(Resistors, Light-emitting diode(LED), wires, buttons)

There will also be required to manufacture a selection of parts for the laparoscope and test station. These parts will mainly consist of 3D printed parts and 1 thin walled rod with a diameter of less than 10 mm.

There will be access required to the mechanical and mechatronic laboratories, to build, program and test the final product.

3.4 Expected Results

The expected results of this project, is to have a low-cost, functional laparoscope at the end of the project. To classify as a low-cost, functional laparoscope, the laparoscope should meet these requirements:

- The laparoscopes rod and camera assembly should fit within a 10 mm diameter hole.

- The captured real-time digital images should allow the operator to accurately coordinate the equipment.
- During testing, the laparoscope should meet a minimum required pass rate of 80%.

3.5 Project Schedule

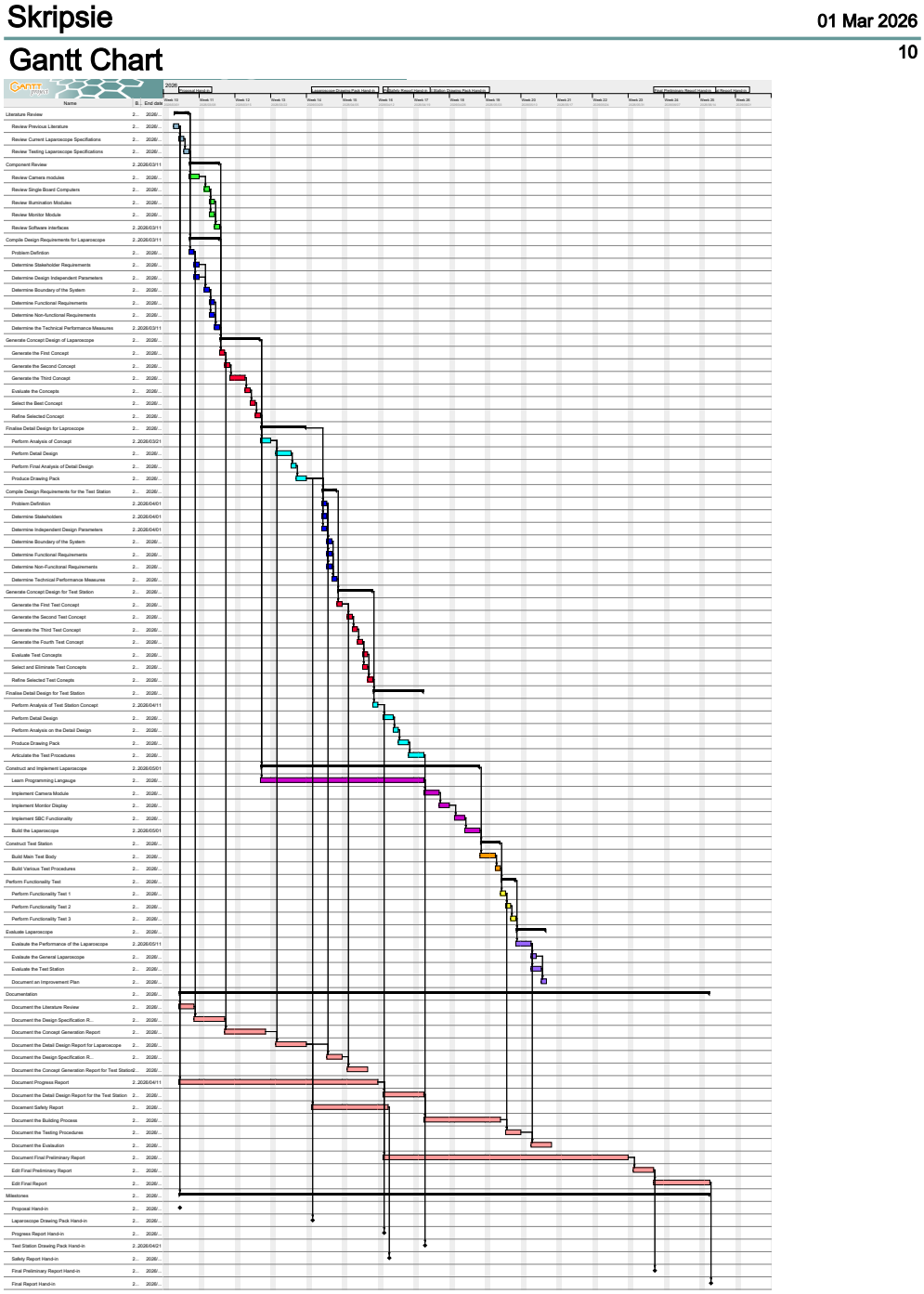

Figure 3.1: Skripsie Gantt Chart

3.6 Risk Assessment

This section of the proposal will cover all relevant risks of this project. Risk is an important feature of any project that needs to be maintained and mitigated to prevent the project from failing.

The first class of risk that will be discussed in this project is that of safety. There are various risk associated with safety problems, for the project and the researcher. These risk mainly occur during building and testing phases. Hence, when the researcher works in the laboratories. These risks include health risks for the researchers and potential damages to the project, leading to time delay to receive replacement parts. To mitigate these risks it will be required to write a safety report, that needs to be reviewed and signed, by the researcher's supervisor and laboratories manager. These safety reports will always be kept on the researcher during laboratories sessions and will required to be completely followed.

Product delay and delivery times is a risk that needs to be handled carefully. Due to the short duration of this skripsie (6 months), product delivery times is a risk to the completion time of this product. If products are required to be imported from international suppliers, then there is a risk that the project could become behind schedule. The only method to mitigate this is by using local suppliers and to order products in a timely manner. If the international products are required, then the products should be order at the earliest convenience. Another mitigation method is to order products local suppliers currently have in stock. It is essential that design critical components are handled with care, to prevent damage to the components, which would require replacement components.

The last class associated with this project is lack of access to the laboratories. This can occur due to several reasons, specifically if there isn't laboratories availability due to limited space or construction of the mechanical and mechatronic department. To minimize this risk, access to the laboratories will be asked at earliest convenience. If access to the mechanical and mechatronic laboratories are unavailable, then access to electrical and electronic departments laboratories will be queried.

Chapter 4

Conclusions

4.1 Conclusion

Laparoscopy is a rapidly expanding industry in the medical field. Its an innovative approach to minimize the harm surgeons inflict on the patients. However, it is a more complex procedure to perform, thus requiring more training. Consequently, there is a desire for medical robots to assist surgeons during laparoscopic surgery.

Therefore, there is a need for a low-cost laparoscope for training purposes. Consequently, it can be used for prototype testing of new medical robots. This low-cost laparoscope is to be tested to determine its functionality. Therefore, a testing station is required to be developed to determine the laparoscope's capabilities.

For this reason, this project aims to develop a low-cost laparoscope in addition to a testing station.

List of references

arducam.com (2020). Arducam 12mp*2 synchronized stereo camera bundle kit for raspberry pi and pi zero, two 12.3mp 477p camera modules with cs lens and camarray stereo camera hat. Tech. Rep., ArduCam.
Available at: https://www.arducam.com/arducam-12mp2-synchronized-stereo-camera-bundle-ki html

Buia, A., Stockhausen, F. and Hanisch, E. (2015). Laparoscopic surgery: A qualified systematic review. *World J Methodol*, vol. 5, no. 4, p. 238–54.

cambridgeincolour.com (2020*a*). Camera exposure. Tech. Rep., Cambridge in Colour.
Available at: https://www.cambridgeincolour.com/tutorials/camera-exposure.htm

cambridgeincolour.com (2020*b*). Digital camera sensors. Tech. Rep., Cambridge in Colour.
Available at: https://www.cambridgeincolour.com/tutorials/camera-sensors.htm

camemake.eu (2026*a*). Endoscope camera module. Tech. Rep., Motoshot Technology Co., Ltd.
Available at: https://www.camemake.com/endoscope-camera-module/

camemake.eu (2026*b*). Home page. Tech. Rep., Motoshot Technology Co., Ltd.
Available at: https://www.camemake.eu/

cameramodule.com (2026). About us. Tech. Rep., Guangzhou Sincere Informtation Technology Ltd.
Available at: https://www.cameramodule.com/about-us

docs.arduino.cc (2026). Language reference. Tech. Rep., Arduino AG.
Available at: https://docs.arduino.cc/language-reference/#structure

ebme.co.uk (2026). Endoscope design and assembly. Tech. Rep., EBME and Clinical Engineering Articles.
Available at: https://www.ebme.co.uk/articles/clinical-engineering/endoscope-design-and-assembly

iadiy.com (2026). Endoscope camera module 0.3m - scm03m30d5.5u. Tech. Rep., IADIY Technology.
Available at: https://www.iadiy.com/endoscope-camera-module-SCM1M30D5.5U

Ibrahim, A., Ulrich, M., Liselotte, M., Julian, P., Nicolai, M., Matthias, B., Georgios, G., Antonio Simone, L. and Damaris, F. (2021). The development of laparoscopy—a historical overview. *Frontiers in surgery*, vol. 8.

intuitive.com (2026). Da vinci sp. Tech. Rep., Intuitive Surgical.
Available at: https://www.intuitive.com/en-us/products-and-services/da-vinci/sp

karlstorz.com (2026*a*). Home page. Tech. Rep., Karl Storz - Endoskope.
Available at: https://www.karlstorz.com/de/en/index.htm

karlstorz.com (2026*b*). Tipcam1 rubina. Tech. Rep., Karl Storz - Endoskope.
Available at: https://www.karlstorz.com/za/en/product-detail-page.htm?productID=1000108799&cat=1000122034&etcc_cu=onsite&etcc_med_onsite=Search-OnlineCatalog-Overlay&etcc_cmp_onsite=Search-Overlay&etcc_st_onsite=Tipcam%20Rubina

leopardimaging.com (2026). Hummingbird mobile camera modules. Tech. Rep., Leopard Imaging.
Available at: https://leopardimaging.com/hummingbird/

mechatronicslab.net (2026). How to use a usb camera with raspberry pi: A complete guide. Tech. Rep., MechatronicsLab.
Available at: https://mechatronicslab.net/usb-camera-with-raspberry-pi/

Morrow, D. (2026). Ultimate depth of field photography guide [2026]. Tech. Rep., Dave Morrow Photography.
Available at: https://www.davemorrowphotography.com/depth-of-field-photography

olympus.com.au (2026). Endoeye 3d 10 mm. Tech. Rep., Olympus Corporation.
Available at: https://www.olympus.com.au/medical/en/Products-and-Solutions/Products/Product/ENDOEYE-3D-10-mm.html

Public (2022). Arducam/arduino. Tech. Rep., Github.com.
Available at: https://github.com/ArduCAM/Arduino

raspberrypi.com (2026*a*). Documentation. Tech. Rep., Raspberry Pi Ltd.
Available at: https://www.raspberrypi.com/documentation/

raspberrypi.com (2026*b*). Products. Tech. Rep., Raspberry Pi Ltd.
Available at: https://www.raspberrypi.com/products/

sincerefirst.com (2026). Sincerefirst camera modules. Tech. Rep., Guangzhou Sincere Information Technology Ltd.
Available at: https://www.sincerefirst.com/camera-modules

store.arduino.cc (2026). Products. Tech. Rep., Arduino AG.
Available at: https://store.arduino.cc/products

Subido, Edwin D. C., J., Pacis, D.M.M. and Bugtai, N.T. (2018). Recent technological advancements in laparoscopic surgical instruments. *AIP Conference Proceedings*, vol. 1933, no. 1.

uk.mathworks.com (2021). Analysis and visualization in thingspeak | iot from data to action, part 3. Tech. Rep., MathWorks.
Available at: https://uk.mathworks.com/videos/iot-from-data-to-action-part-3-analysis-and-visualization-in-thingspeak-1636518392833 html

Xiao, D., Jakimowicz, J.J., Albayrak, A., Buzink, S.N., Botden, S.M.B.I. and Goossens, R.H.M. (2014). Face, content, and construct validity of a novel portable ergonomic simulator for basic laparoscopic skills. *Journal of surgical education*, vol. 71, no. 1, p. 65–72.

Yu, J. (2026). Latex workshop. Tech. Rep., Visual Studio.
Available at: https://marketplace.visualstudio.com/items?itemName=James-Yu.latex-workshop

Estimated Cost per Activity

The estimated cost per activity, shows all the planned activities of the project and how much time would be used to perform each activity. Each activity duration is determine by the gannt chart. It is assumed that there is 8 hours available each working day. And the workload is divided based on importance for each day. Usaully 5 hours for planned activities and then 1 hour each day for documentation. This is not a concrete value and is subject to change.

Table .1: Estimated Cost per Activity

Activity	Engineering Time		Running Cost	Facility Use	Capital Cost	MMW			Total
						Labour		Material	
	hr	R	R	R	R	hr	R	R	R
Literature Review	15	6750	150						6900
Component Review	20	9000	150						9150
Compile Design Requirements for Laparoscope	20	9000	150						9150
Generate Concept Design for Laparoscope	35	15750	150						15900
Finalise Detail Design for Laparoscope	40	18000	150						18150
Compile Design Requirements for Test Station	20	9000	150						9150
Generate Concept Design for Test Station	30	13500	150						13650
Finalise Detail Design for Test Station	40	18000	150						18150
Constuct and Implement Laparoscope	95	42750	600	6000	4000	15	4500	500	58350
Construct Test Station	20	9000	600	2000		40	12000	300	23900
Perform Functionality Test	40	18000	600	2000					20600
Evaluate Laparoscope	25	11250	150						11400
Documentation	120	54000	150						54150
Total	**520**	**234000**	**3300**	**10000**	**4000**	**55**	**16500**	**800**	**268600**

GA Self Assessment

Graduate Attribute	How it will be satisfied
GA 1 - Problem solving: Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development.	This GA will be satisfied during the literature review, design specification reports, concept generation reports and the detail design reports, including drawing packs.
GA 2 - Application of scientific and engineering knowledge: Apply knowledge of mathematics, natural science, computing and engineering fundamentals, and an engineering specialisation to develop solutions to complex engineering problems.	GA 2 will be satisfied during sections where component specification and selection will be conducted. It will also be satisfied during detail design sections where analysis of the design will be performed.
GA 3 - Engineering Design: Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs.	This GA will be satisfied by following the workplan and planned schedule of the project. It will also be satisfied during component selection, detail design process and the evaluation of the developed product.

Continued on next page...

Graduate Attribute (continued)	How it will be satisfied (continued)
GA 5 - Use of Engineering Tools: Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems.	GA 5 will be satisfied during the detail design of the project. It will also be satisfied during the implementation of the laparoscope.
GA 6 - Professional and technical communication: Demonstrate competence to communicate effectively and inclusively on complex engineering activities, both orally and in writing, with the engineering community and society at large, taking into account cultural, language, and learning differences.	GA 6 will be satisfied during the documentation of the project, including the progress and final report. It will also be satisfied during the weekly meetings, progress meeting and final presentation.
GA 8 - Individual and collaborative teamwork: Demonstrate competence to function effectively as an individual and as a member or leader in diverse and inclusive teams, and in multidisciplinary, face-to-face, remote and distributed settings.	This GA will be satisfied during the construction of the laparoscope and testing station. It will also be satisfied during the testing stages of the laparoscope.
GA 9 - Independent Learning Ability: Demonstrate competence to engage in independent learning through well-developed learning skills.	This will be satisfied during the entirety of the project, but predominantly during the literature review.

Table .2: GA self-assessment